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MSEI-023

**MASTER OF SCIENCE
(INFORMATION SECURITY)/
P. G. DIPLOMA IN INFORMATION
SECURITY
(MSCIS/PGDIS)**

Term-End Examination

June, 2023

MSEI-023 : CYBER SECURITY

Time : 2 Hours

Maximum Marks : 50

Note : (i) **Section A :** Answer all the questions.

(ii) **Section B :** Answer any **two** out of three questions.

(iii) **Section C :** Answer any **two** out of three long answer type questions.

P. T. O.

Section—A

Note : Attempt all the questions.

1. Write notes on the following (any *four*) :

5×4=20

- (a) Client / Server architecture
- (b) Social Engineering
- (c) Cluster table
- (d) Social Networking sites
- (e) Trojan Horse

Section—B

Note : Attempt any *two* out of *three* questions.

- 2. Describe the architecture of Oracle Database Management System. 5
- 3. Describe SQL injection attack with the help of a suitable example. 5
- 4. What is phishing attack ? How do you identify a phishing e-mail ? 5

Section—C

Note : *Attempt **two** out of three long answer type questions.*

5. Explain the security issues involved in centralised as well as distributed database systems. Which one is more insecure and why ? 10
6. Describe Cross Site Scripting (XSS) and Cross Site Request Forgery (XSRF) attacks and steps to thwart/avoid such attacks. 10
7. Explain the shadow paging recovery technique. 10

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Time : 2 Hours

Maximum Marks : 50

Note : *Question No. 1 is compulsory. Attempt any
three questions from the rest.*

1. Write short notes on the following : 4×5=20
 - (a) Principle of least privilege in access control
 - (b) Zero-Day Vulnerabilities
 - (c) Role of Multi-Factor Authentication (MFA)
 - (d) Social Engineering attacks

2. What is domain cache poisoning attack ? How
can it be prevented ? 10

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P. T. O.

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3. Explain the architecture and working of a Public Key Infrastructure (PKI). 10
4. Describe the steps involved in performing a Vulnerability Assessment and Penetration Testing (VAPT) on a network. 10
5. Describe the lifecycle of a ransomware attack, from initial infection to encryption and ransom demands. 10

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- Note :** (i) **Section A :** Answer all the questions.
(ii) **Section B :** Answer any **two** out of three questions.
(iii) **Section C :** Answer any **two** out of three long answer type questions.
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Section—A

1. Write notes on any *four* of the following :

4×5=20

- (a) Trojan Horses
- (b) Hybrid Crack
- (c) Role of scripting language in the cyber world
- (d) Entity-Relationship (ER) model
- (e) Social bookmarking

P. T. O.

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Section—B

Note : Attempt any *two* out of *three* questions.

2. Describe the process of using e-mail location tracking tool. 5
3. Explain SQL injection attack with the help of a suitable example. 5
4. Discuss social engineering attack cycle. 5

Section—C

Note : Attempt two out of three long answer type questions.

5. What is Concurrency Control in a DBMS ? What are the methods of concurrency control in a DBMS ? Also discuss the goals of concurrency control mechanism. 10
6. List at least *five* types of cyber attacks. Explain *two* of them in detail including the preventive measures that should be adopted to thwart such attacks. 10
7. Explain the three tier client server architecture with the help of a diagram. Discuss the advantages of three-tier client server architecture over two-tier client server architecture. 10

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Time : 2 Hours

Maximum Marks : 50

Note : *Question No. 1 is compulsory. Attempt any **three** questions from the rest.*

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1. Write short notes on the following : 4×5=20
 - (a) OWASP top ten vulnerabilities for web application security.
 - (b) Advanced Persistent Threats (APT's)
 - (c) Botnets
 - (d) Cloud security and its unique challenges

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2. How are passwords stored at a server (e. g. at an e-Commerce establishment) ? How can hackers get user-passwords by stealing information from the servers ? 10
3. Explain the key principles and components of the Kerberos authentication protocol. Provide a step-by-step description of how a user can authenticate to a service using Kerberos. 10
4. Explain the process of a man-in-the-middle (MITM) attack and various techniques used to execute it. 10
5. Describe the steps involved in security a cloud computing environment, focusing on Identity and Access Management (IAM), encryption and network security. 10

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December, 2025

MSEI-023 : CYBER SECURITY

Time : 2 Hours

Maximum Marks : 50

Note : (i) *Question No. 1 is compulsory.*

(ii) *Attempt any **three** questions from
Q. Nos. 2 to 5.*

1. Write short notes on the following : 4×5=20

- (i) Types of Malware (e.g. viruses, worms,
trojans)

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- (ii) Importance of VPNs in network security
 - (iii) Concept of Zero-Day Vulnerabilities
with examples
 - (iv) Concept of Zero Trust in Information
Security
2. Explain the principle of least privilege in access control systems. How does it enhance cyber security ? 10
3. Discuss the steps involved in conducting a Vulnerability Assessment and Penetration Testing (VAPT). Provide examples where VAPT is critical for organisations. 10
4. Describe the lifecycle of a ransomware attacks from initial infection to encryption and ransom demands. What measures can organisations take to prevent such attacks ? 10

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5. What are social engineering attacks ?

Explain its different types with examples and discuss how individuals and organizations can mitigate such attacks effectively.

10

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**MASTER OF SCIENCE
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Term-End Examination**

June, 2026

MSEI-023 : CYBER SECURITY

Time : 2 Hours

Maximum Marks : 50

Note : (i) *Question No. 1 is compulsory.*

(ii) *Attempt any **three** questions from*

Q. 2 to Q. 5.

1. Write short notes on the following : 4×5=20

- (i) Difference between IDS and IPS with examples.

- (ii) Role of Multi-factor Authentication (MFA) in securing systems.
 - (iii) Phishing *vs.* Spear Phishing attacks with example.
 - (iv) Steps to prevent domain Cache poisoning attacks.
2. Explain Information security in the context of Cloud Infrastructure. How does it ensure secure communication over the Internet ? 10
3. Discuss how firewalls and intrusion detection systems (IDS) help secure networks from cyber threats. Provide examples of their application in real-world scenarios. 10
4. What are Zero-Day vulnerabilities ? Explain their significance in cyber-security with examples of recent incidents involving zero-day exploits. 10

[3]

5. Describe the importance of conducting regular cyber security audit for organizations to identify vulnerabilities and mitigate risks effectively. Name a commonly referred International standard and provide the steps involved in such audits. 10

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